

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform (FinRelief AI)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed standard Python, React.js, and JavaScript formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Project separated into frontend, backend, database, and API modules.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used descriptive names such as loan_amount, financial_score, and settlement_prediction.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly represent their purpose (e.g., generateSettlement(), analyzeFinancialHealth()).
5	Code Comments	Inline and block comments explain logic	Partial	Comments added for important modules and complex AI logic
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules created for authentication, loan management, AI processing, and database operations.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common functionality reused through modular functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Implemented API validation, exception handling, and user-friendly error messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	FastAPI, Python	User login and registration	High

2	Financial Analysis Module	Python	Financial health calculations and debt analysis	High
3	AI Negotiation Module	Google Gemini API, Python	Negotiation strategy and settlement letter generation	High
4	Loan Management Module	FastAPI, SQLite	Loan creation, updates, and settlement records	High
5	Dashboard Components	React.js	Financial dashboard and analytics pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular architecture with separate frontend and backend layers.
Readability	5	Descriptive naming conventions and consistent formatting improve readability.
Reusability	5	Reusable modules reduce code duplication and simplify maintenance.
Documentation / Comments	4	Key modules are documented with comments where necessary.
Overall Score	4.8	Clean, modular, maintainable, and scalable code following good software engineering practices.